PROFESSIONAL PROGRAM | COURSE OVERVIEW

Diploma in Data Science with AI

From Python fundamentals to Generative AI, deployment and career readiness.

7 MODULES • BEGINNER TO JOB-READY

Program Overview

The Diploma in Data Science with AI is a structured, hands-on program that takes you from the basics of programming and statistics to building and deploying modern AI systems, including deep learning and generative AI. It is designed for beginners and career switchers aiming for roles such as Data Analyst, Data Scientist, ML Engineer and AI Developer.

Modules at a Glance

#	Module	Key Topics
1	**Python and Statistics Foundation**	Python, NumPy, pandas, probability, inference
2	**Databases and Big Data**	SQL, NoSQL, data warehouses, Spark basics
3	**Machine Learning Core**	Supervised and unsupervised models, feature engineering
4	**Deep Learning**	Neural networks, CNNs, RNNs, transformers with PyTorch
5	**NLP and Generative AI**	Embeddings, LLM apps, retrieval, prompt design
6	**MLOps and Cloud**	Docker, APIs, model monitoring, cloud basics
7	**Capstone and Placement Prep**	Industry project, resume, interview practice

Learning Path

Foundation (1) → Data Systems (2) → Machine Learning (3) → Deep Learning (4) → Language and Generative AI (5) → Deployment (6) → Career (7)

Each module builds on the previous one, so skills compound as you progress.

1 Python and Statistics Foundation

Python, NumPy, pandas, probability, inference

This module builds the base for everything that follows. You learn to write Python and use it to handle data, while learning the statistics needed to make sense of it.

What you will learn

- **Python:** A beginner-friendly programming language used widely in data work and AI.
- **NumPy:** A library for fast numerical computing with arrays and matrices.
- **pandas:** A library for loading, cleaning, filtering and summarising tabular data.
- **Probability:** The mathematics of chance; it measures how likely events are.
- **Inference:** Drawing conclusions about a large population from a smaller sample, using tools like hypothesis tests and confidence intervals.

> **Outcome:** Clean and analyse a real dataset and report findings with statistical backing.

2 Databases and Big Data

SQL, NoSQL, data warehouses, Spark basics

Data lives in databases, not spreadsheets. This module teaches how to store, query and process data at small and large scale.

What you will learn

- **SQL:** The standard language for querying relational (table-based) databases.
- **NoSQL:** Flexible databases (document, key-value, graph) for unstructured or fast-changing data.
- **Data warehouses:** Central stores optimised for analytics, combining data from many sources.
- **Spark basics:** A distributed engine that processes very large datasets across many machines.

Outcome: Write queries to extract insights and run a simple large-scale processing job.

3 Machine Learning Core

Supervised and unsupervised models, feature engineering

Machine learning lets computers learn patterns from data instead of following hand-written rules. This is the heart of the diploma.

What you will learn

- **Supervised learning:** Models trained on labelled examples, such as predicting prices (regression) or spam (classification).
- **Unsupervised learning:** Finding hidden structure without labels, such as customer clustering.
- **Feature engineering:** Creating and selecting the input variables that help a model perform better.
- **Evaluation:** Measuring models with metrics and validation to avoid overfitting.

Outcome: Build, tune and compare models using scikit-learn.

4 Deep Learning

Neural networks, CNNs, RNNs, transformers with PyTorch

Deep learning uses layered neural networks to solve complex problems like image recognition, speech and language.

What you will learn

- **Neural networks:** Layers of connected units that learn by adjusting weights from data.
- **CNNs:** Convolutional networks, specialised for images and visual data.
- **RNNs:** Recurrent networks, designed for sequences such as time series and text.
- **Transformers:** The attention-based architecture behind modern language models.
- **PyTorch:** A popular framework for building and training deep learning models.

Outcome: Train an image classifier and a sequence model in PyTorch.

5 NLP and Generative AI

Embeddings, LLM apps, retrieval, prompt design

This module covers how machines understand language and how to build applications powered by large language models (LLMs).

What you will learn

- **Embeddings:** Numeric representations of text that capture meaning, so similar ideas sit close together.
- **LLM apps:** Applications such as chatbots and assistants built on top of language models.
- **Retrieval (RAG):** Fetching relevant documents and giving them to an LLM so answers are grounded in real data.
- **Prompt design:** Writing clear instructions and examples to get reliable model outputs.

> **Outcome:** Build a document question-answering assistant using retrieval.

6 MLOps and Cloud

Docker, APIs, model monitoring, cloud basics

A model is only useful when it runs reliably in the real world. MLOps covers deploying, serving and maintaining models.

What you will learn

- **Docker:** Packages code and dependencies into containers that run the same everywhere.
- **APIs:** Expose a model as a web service that other applications can call.
- **Model monitoring:** Tracking accuracy, drift and performance after deployment.
- **Cloud basics:** Using services like AWS, Azure or GCP for storage, compute and hosting.

> **Outcome:** Containerise a model, serve it through an API and deploy it to the cloud.

7 Capstone and Placement Prep

Industry project, resume, interview practice

The final module brings everything together and prepares you for the job market.

What you will learn

- **Industry project:** An end-to-end project: define the problem, build, deploy and present.
- **Resume:** Building a clear resume and portfolio that showcase your projects.
- **Interview practice:** Mock interviews covering technical questions, case studies and communication.

Outcome: A portfolio-ready project and interview confidence.